An explicit tenth-discriminant bound
Version 2: the refined cutoff

Research note prepared with Codex

Version 0.2.0 — 20 October 2026

Abstract

Taking explicitly identified results of Banks (`arXiv:2610.15483v1`) as external inputs, we obtain $q < 5{\cdot}10^7$ for imaginary quadratic fields of class number one. This self-contained cumulative version retains the original $q < 10^{10}$ argument, proves the refinement, and includes the finite exclusion that leaves exactly the nine familiar fundamental discriminants. The refinement changes both the theta estimate and the parameters in Banks's determinant argument. Neither version uses Heegner's methods or Banks's separate small-conductor zero computation.

1 What changed since version 1

Version 1 remains available unchanged, including its source, PDF, programs, and recorded outputs. This version contains the whole argument needed to read and reproduce the refined result.

- The analytic cutoff improves from 10^{10} to $5 \cdot 10^7$, a factor of 200.
- The original theta estimate $1 - \beta < 10/\sqrt{q}$ is retained in Section 3. The refinement proves $1 - \beta < 2/\sqrt{q}$ for $q \geq 49\,000\,000$.
- The original argument uses Banks's stated Theorem 1.1. The refinement uses Lemma 4.2, Lemma 5.1, and Corollary 6.3, including their preceding construction and hypotheses, with $H = 10^9$ and $N = q^{18}$. Thus the change is not a substitution of a smaller threshold into Theorem 1.1.
- The two original integer sieve programs and their outputs are preserved byte-for-byte. Their search below 10^{10} also covers the smaller remaining interval. A new exact rational certificate checks the constants used in the refinement.

The repository owner reports a cross-check of the original mathematical argument with a different frontier model. The new refinement has not yet received that separate-model cross-check. Its numerical inequalities have passed the exact arithmetic certificate. These are distinct verification records, detailed in Section 8.

2 Results and external dependencies

Throughout, $D < 0$ is a fundamental discriminant, $q = |D|$, $K = \mathbb{Q}(\sqrt{D})$, and χ_D is the primitive quadratic character of conductor q. All logarithms are natural. The quantity q is the absolute field discriminant, not necessarily the squarefree integer under the square root.

Theorem 1 (Refined bound)**.** *Assume Lemma 4.2, Lemma 5.1, and Corollary 6.3 of Banks [1], including their preceding construction and hypotheses. If $h(K) = 1$, then*

$$q < 5 \cdot 10^7.$$

Together with the finite verification in Section 7, this gives

$$q \in \{3, 4, 7, 8, 11, 19, 43, 67, 163\}.$$

Every discriminant on this list does have class number one.

For comparison, the original external input was Banks's Theorem 1.1:

$$q \geq 10^{10}, \quad L(\beta, \chi) = 0, \quad 0 < \beta < 1 \quad \Longrightarrow \quad (1 - \beta) \log q \geq \frac{3}{1000} \tag{1}$$

for every primitive real character χ of conductor q.

Proposition 2 (Original bound)**.** *Assuming (1), if $h(K) = 1$, then $q < 10^{10}$.*

The external assumptions are stated explicitly because this note does not independently prove Banks's theorem or the intermediate lemmas. The contribution is their explicit application to the class-number-one problem, with reproducible finite exclusion. No novelty claim is made for the classical theta method, and no optimality claim is made for the numerical cutoff.

3 Original argument: the cutoff 10^{10}

Lemma 3. *If $q \geq 10^{10}$ and $h(K) = 1$, then $L(s, \chi_D)$ has a real zero*

$$1 - \frac{10}{\sqrt{q}} < \beta < 1.$$

Proof. Put

$$R = \sqrt{q}, \qquad y = \frac{R}{2}, \qquad \Lambda = y^{-1/2} \mathcal{O}_K \subset \mathbb{C} \simeq \mathbb{R}^2.$$

The embedded integer lattice $\mathcal{O}_K$ has covolume y, so Λ has covolume one. Define

$$\Theta(t) = \sum_{\alpha \in \mathcal{O}_K} \exp\left(-\frac{\pi t |\alpha|^2}{y}\right) \qquad (t > 0).$$

For any planar lattice of covolume one, its Euclidean dual is a rotation by 90 degrees of the original lattice. For example, if the columns of A are an oriented basis and $\det A = 1$, then $A^{-T} = JAJ^{-1}$, where J is rotation by 90 degrees and $J^{-1}\mathbb{Z}^2 = \mathbb{Z}^2$. Consequently, Poisson summation for the Gaussian gives

$$\Theta(t) = t^{-1} \Theta(1/t). \tag{2}$$

Since $q > 4$, the units of $\mathcal{O}_K$ are exactly $\{\pm 1\}$. Class number one implies that every nonzero integral ideal has exactly two generators in $\mathcal{O}_K$. Thus, initially for $\Re s > 1$,

$$\sum_{\alpha \in \mathcal{O}_K \setminus \{0\}} |\alpha|^{-2s} = 2\zeta_K(s).$$

Taking the Mellin transform and splitting its integral at $t = 1$, then using (2) in the part below 1, gives

$$2\Gamma(s)\left(\frac{y}{\pi}\right)^s \zeta_K(s) = -\frac{1}{s(1-s)} + \int_1^\infty (\Theta(t)-1)\left(t^{s-1}+t^{-s}\right)\,dt. \tag{3}$$

The integral on the right converges locally uniformly for all complex s, since $\Theta(t) - 1$ decays exponentially as $t \to \infty$. Hence (3) continues the initial identity meromorphically. For clarity, the rational term is

$$\int_0^1 (t^{s-2} - t^{s-1})\,dt = \frac{1}{s-1} - \frac{1}{s} = -\frac{1}{s(1-s)}.$$

Set

$$\varepsilon = \frac{10}{R}, \qquad s_0 = 1-\varepsilon, \qquad A = \frac{y}{\pi} = \frac{R}{2\pi}.$$

Here $R \geq 10^5$, $0 < s_0 < 1$, and $A > 1$. The two lattice points $\alpha = 1, -1$ alone give

$$\Theta(t) - 1 \geq 2e^{-\pi t/y} \geq \frac{2}{e} \qquad (1 \leq t \leq A).$$

Dropping the nonnegative t^{-s_0} term and the part of the integral above A, the integral in (3) is therefore at least

$$\frac{2}{e}\int_1^A t^{s_0-1}\,dt = \frac{2}{e\,s_0}(A^{s_0}-1).$$

It suffices to prove

$$\frac{2\varepsilon}{e}(A^{1-\varepsilon}-1) > 1. \tag{4}$$

All estimates needed for (4) can be made rational. The function $(\log R)/R$ decreases for $R > e$, and $\log(10^5) < 12$. Hence

$$0 < \varepsilon \log A \leq \frac{10\log R}{R} < \frac{3}{2500}.$$

Using $e^{-u} \geq 1 - u$,

$$A^{1-\varepsilon} = Ae^{-\varepsilon \log A} > \frac{R}{2\pi}\frac{2497}{2500}.$$

The elementary bounds $\pi < 22/7$ and $2 < e < 3$ now imply

$$\begin{aligned}\frac{2\varepsilon}{e}(A^{1-\varepsilon}-1) &> \frac{10}{\pi e}\frac{2497}{2500} - \frac{20}{eR} \\ &> \frac{35}{33}\frac{2497}{2500} - \frac{1}{10000} \\ &= \frac{31777}{30000} > 1.\end{aligned}$$

For an entirely rational check of the logarithmic bound, use $e > 8/3$ and $8^{12} > 10^5\,3^{12}$. This proves (4), and so $\zeta_K(s_0) > 0$.

As $s \to 1^-$, the rational term in (3) tends to $-\infty$, while its integral stays finite. Thus $\zeta_K(s) < 0$ sufficiently close to 1 from the left. Continuity gives a zero $\beta \in (s_0, 1)$. Finally $\zeta_K(s) = \zeta(s)L(s,\chi_D)$, and $\zeta(s) \neq 0$ for real $0 < s < 1$. One elementary proof of the latter uses the positive alternating series

$$\eta(s) = \sum_{n\geq 1}(-1)^{n-1}n^{-s} > 0, \qquad \zeta(s) = \frac{\eta(s)}{1-2^{1-s}} < 0.$$

Therefore $L(\beta, \chi_D) = 0$, as claimed. □

Proof of Proposition 2. If $q \geq 10^{10}$ and $h(K) = 1$, Lemma 3 gives

$$(1-\beta)\log q < \frac{10\log q}{\sqrt{q}}.$$

The function $(\log q)/\sqrt{q}$ decreases for $q > e^2$. Using $\log(10^{10}) < 24$, we obtain

$$(1-\beta)\log q < \frac{10 \cdot 24}{10^5} = \frac{3}{1250} < \frac{3}{1000}.$$

This contradicts (1). □

4 Refinement A: a stronger zero-free estimate

We first show that those lemmas imply the following numerical statement: for every primitive real character of conductor $q \geq 5 \cdot 10^7$, every real zero $0 < \beta < 1$ satisfies

$$\delta := (1-\beta)\log q > \frac{509}{100000} = 0.00509. \tag{5}$$

Put

$$H = 10^9, \qquad N = q^{18}, \qquad M = N^{4/3} = q^{24}, \qquad L = \log q.$$

Take $\lambda = 3/2$ in Banks's prime-bias input. Its constants satisfy

$$C_1 = \frac{2e^{3/2}}{9} < 1, \qquad C_2 = \frac{e^{3/2}}{4} < \frac{9}{8}.$$

The hypotheses $q \geq 9$, $319 \leq H \leq N$, $M \geq 8$, and $H^{2/3}N^{4/3} \geq 2 \cdot 97^2$ all hold. In particular, Banks's construction requires $H \leq N$, not $H \leq q$. The latter was a sufficient condition used with his original parameters, and is not needed here.

Use the nonzero determinant and the degree sums S_1, S_{23} in Banks's notation. Lemma 4.2 gives

$$S_1 \geq \frac{10^6}{4 \cdot 97^2} N^4 M > 25 N^4 M, \qquad \frac{S_{23}}{S_1} \leq \frac{768 \cdot 97^2}{10^9} < \frac{1}{135}.$$

Combining Lemma 5.1 with Corollary 6.3, dividing by $S_1 > 0$, and using $\log N = \frac{3}{4}\log M$, gives

$$\frac{11}{45}\log M \leq \delta\frac{(\log M)^2}{L} + \left(\frac{9}{8} + \frac{1}{2H} + \frac{68}{135}\right)L + \log H + \frac{2}{25} + \frac{136}{135}\log 8 + \frac{2\log N}{25M}. \tag{6}$$

Here $11/45 = 1 - (136/135)(3/4)$. Replacing C_1 by 1 increases the right side because $\delta > 0$.

The coefficient of L after the first term is less than $163/100$. Also

$$\log H < 9(2.303), \qquad \log 8 < 2.08, \qquad \frac{2\log N}{25M} < \frac{1}{1000},$$

so the remaining terms on the right of (6) sum to less than 23. For the last inequality, $N \geq 1000$ and the function $(\log N)/N^{4/3}$ is decreasing there; use $\log 1000 < 7$. Since $\log M = 24L$, (6) implies

$$576\delta > \frac{88}{15} - \frac{163}{100} - \frac{23}{L}.$$

Finally $L \geq \log(5 \cdot 10^7) > 17.7$, and hence

$$\delta > \frac{1}{576}\left(\frac{88}{15} - \frac{163}{100} - \frac{230}{177}\right) > \frac{509}{100000}. \tag{7}$$

This proves (5), without a computation of any Dirichlet zeros.

5 Refinement B: a closer real zero

Lemma 4. *If $q \geq 49\,000\,000$ and the class number is one, then there is a real zero $0 < \beta < 1$ of $L(s, \chi_{-q})$ such that*

$$1 - \beta < \frac{2}{\sqrt{q}}.$$

Proof. Use the theta function and Mellin identity (3) derived in Section 3, with $R = \sqrt{q}$ and $y = R/2$. That identity requires only $q > 4$ and class number one; the stronger range restriction there was used for the subsequent numerical estimate. Here $R \geq 7000$.

Set

$$\varepsilon = \frac{2}{R}, \qquad s = 1 - \varepsilon, \qquad A = \frac{R}{2\pi}, \qquad T = 5A.$$

Retain just the lattice points $\alpha = \pm 1, \ldots, \pm 30$ and just the t^{s-1} term over $1 \leq t \leq T$. If I denotes the integral in (3), then

$$\begin{aligned} I &\geq 2T^{-\varepsilon} \sum_{n=1}^{30} \int_1^T e^{-n^2 t/A}\, dt \\ &= 2AT^{-\varepsilon} \sum_{n=1}^{30} \frac{e^{-n^2/A} - e^{-5n^2}}{n^2} \\ &\geq 2AT^{-\varepsilon} \left(\sum_{n=1}^{30} \frac{1}{n^2} - \frac{30}{A} - \sum_{n=1}^{30} \frac{e^{-5n^2}}{n^2} \right). \end{aligned}$$

The last step uses $e^{-u} \geq 1 - u$. For explicit bounds, use

$$\sum_{n=1}^{30} \frac{1}{n^2} > \frac{1612}{1000}, \qquad \sum_{n=1}^{30} \frac{e^{-5n^2}}{n^2} \leq \frac{e^{-5}}{1 - e^{-15}} < \frac{7}{1000}.$$

Indeed $n^2 \geq 1 + 3(n-1)$; and the second strict inequality follows already from $e > 19/7$. Since $R \geq 7000$, $3 < \pi < 22/7$, and $(\log R)/R$ decreases,

$$\frac{30}{A} < \frac{1320}{49000}, \qquad s \geq 1 - \frac{2}{7000}, \qquad T^{-\varepsilon} \geq 1 - \varepsilon \log T > 1 - \frac{18}{7000}.$$

For the last inequality, $T < R$ and $\log 7000 < 9$. It follows that

$$\begin{aligned} \varepsilon s I &> \frac{7}{11} \left(1 - \frac{2}{7000}\right) \left(1 - \frac{18}{7000}\right) \left(\frac{1612}{1000} - \frac{1320}{49000} - \frac{7}{1000}\right) \\ &= \frac{37781022837}{37730000000} > 1. \end{aligned}$$

Thus $\zeta_K(1 - 2/R) > 0$ by (3). The same identity shows that $\zeta_K(s) \to -\infty$ as $s \to 1^-$, so continuity gives a zero $\beta \in (1 - 2/R, 1)$. Since $\zeta_K(s) = \zeta(s) L(s, \chi_{-q})$ and $\zeta(s) < 0$ for real $0 < s < 1$, this is a zero of the Dirichlet L-function. □

6 Conclusion of the refined bound

If $q \geq 5 \cdot 10^7$ had class number one, Lemma 4 would give

$$(1 - \beta) \log q < \frac{2 \log q}{\sqrt{q}}.$$

The function on the right decreases for $q > e^2$. Using $\log(5 \cdot 10^7) < 17.73$ and $\sqrt{5 \cdot 10^7} > 7071$, we therefore get

$$(1-\beta)\log q < \frac{2(17.73)}{7071} < \frac{502}{100000} = 0.00502.$$

This contradicts (5) and proves Theorem 1.

7 Finite exclusion: the original certificate

The analytic refinement leaves only $q < 5 \cdot 10^7$. We retain the original programs and their recorded search over the larger interval $q < 10^{10}$, so the verification below covers every remaining case.

Lemma 5. *If $h(K) = 1$, then every prime $p < q/4$ is inert in K.*

Proof. If p splits or ramifies, there is an integral ideal of norm p. It is principal when $h(K) = 1$, so it has a generator $\alpha \in \mathcal{O}_K$ with $|\alpha|^2 = p$. A rational integer cannot have prime norm in a quadratic field. Every nonrational algebraic integer in K can be written $(a + b\sqrt{-q})/2$, with integers a, b and $b \neq 0$; the allowed parities depend on D. Consequently,

$$p = |\alpha|^2 = \frac{a^2 + qb^2}{4} \geq \frac{q}{4},$$

a contradiction. □

For $q > 8$, Lemma 5 applies to $p = 2$. Thus 2 is inert, D is odd, and $D \equiv 5 \pmod 8$; equivalently, $q \equiv 3 \pmod 8$. The negative fundamental discriminants with $q \leq 8$ are $-3, -4, -7, -8$.

For every remaining candidate, therefore, it is enough to examine

$$11 \leq q < 10^{10}, \qquad q \equiv 3 \pmod 8.$$

For an odd prime $p < q/4$, exclude q whenever $-q$ is zero or a square modulo p. The search deliberately includes nonfundamental q, so no squarefreeness test can accidentally discard a valid candidate.

Exact computation

Two programs accompany this note:

- `verify_sieve.cpp` builds permitted residue classes using the Chinese remainder theorem, then checks candidates individually using tables of squares modulo primes.
- `verify_independent.py` obtains the initial residue classes by direct scanning and Euler's criterion. It then checks whole arithmetic progressions using periodic integer bitsets.

Both first impose the conditions for $2, 3, 5, 7, 11, 13, 17$. The resulting wheel has modulus

$$M = 8 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13 \cdot 17 = 2042040$$

and 1440 permitted residue classes. For $437 \leq q < 10^{10}$, every prime at most 109 satisfies $4p < q$. The programs check the remaining primes from 19 through 109, and find no surviving q. For $11 \leq q < 437$, they apply only the prime conditions with $4p < q$, and find exactly

$$11, \quad 19, \quad 43, \quad 67, \quad 163.$$

Both programs retain the original exclusive upper endpoint 10^{10}. The refined analytic argument already excludes every $q \geq 5 \cdot 10^7$. These calculations use exact integer arithmetic throughout.

Checking the surviving fields

The Python verifier independently enumerates all primitive reduced positive binary quadratic forms of each remaining discriminant. The unique reduced form in each case is as follows:

q	3	4	7	8	11	19	43	67	163
(a,b,c)	$(1,1,1)$	$(1,0,1)$	$(1,1,2)$	$(1,0,2)$	$(1,1,3)$	$(1,1,5)$	$(1,1,11)$	$(1,1,17)$	$(1,1,41)$

In the enumeration, $|b| \le a \le c$, $b^2 - 4ac = -q$, $\gcd(a,b,c) = 1$, and $b \ge 0$ on the boundaries $|b| = a$ or $a = c$. The standard bound $a \le \sqrt{q/3}$ makes the enumeration finite. The classical correspondence with ideal classes verifies class number one for these nine fields.

8 Reproduction, scope, and verification

Run the following commands from this version's directory:

```
g++ -O2 -std=c++17 -Wall -Wextra -pedantic \
    verify_sieve.cpp -o verify_sieve
./verify_sieve
python3 verify_independent.py
python3 verify_improved_bound.py
```

The programs require no number-theory library or external data. The original sieve outputs are `sieve_cpp_output.txt` and `sieve_python_output.txt`; they are reused verbatim from version 1. The two algorithms agree, and reduced-form enumeration checks all nine survivors. The original Python verifier also checks the rational inequalities used in the original analytic estimate.

The new certificate, `verify_improved_bound.py`, checks 23 finite inequalities and identities using exact integers and rational numbers. Its output is `improved_bound_checks.txt`. For logarithms and exponentials it uses rational Taylor bounds: if $x \ge 0$, $n + 2 > x$, and $S_n(x) = \sum_{k=0}^{n} x^k/k!$, then

$$S_n(x) \le e^x \le S_n(x) + \frac{x^{n+1}}{(n+1)!}\frac{1}{1 - x/(n+2)}.$$

The tail bound follows by bounding successive term ratios by $x/(n+2) < 1$. Monotonicity of the exponential converts these to certificates for the logarithmic inequalities. The standard bounds $3 < \pi < 22/7$ are mathematical inputs.

These numerical checks do not constitute formal verification of the analytic deductions or of Banks's lemmas. The classical inputs include Poisson summation for a Gaussian, quadratic ideal theory, $\zeta_K = \zeta L$, and reduction of positive binary quadratic forms. No Lean formalization is supplied. Banks's separate small-conductor zero computation is not used.

Version-specific review status. The repository owner reports that the original version's mathematical argument was cross-checked with a different frontier model. No detailed report of that cross-check is bundled here. The refinement in Sections 4–5 has not yet received a separate-model mathematical review. Its arithmetic certificate passes, and the typeset paper has been rendered and visually inspected. The repository's `REVIEW.md` records the scope of these checks.

AI disclosure and attribution. Codex produced the route selection, explicit estimates, proof text, and all three verification programs. The two sieve algorithms were written by the same assistant. Their agreement checks computation; it does not supply an independent review of the analytic proof. The external determinant results are due to the cited Banks manuscript, which attributes its underlying argument to the OpenAI manuscript. This contribution is prepared for Tao's collaborative challenge [2]. It does not claim an independent proof of Banks's results.

The original release and this cumulative version have separate sources, PDFs, checksums, and fixed version references. Source provenance is recorded in `sources/manifest.json`. The inherited provenance also records a Green manuscript consulted during the original research; it is not an input to either proof here.

References

[1] W. D. Banks, *On the OpenAI proof that Landau–Siegel zeros do not exist*, 11 October 2026, arXiv:2610.15483v1. Theorem 1.1 for the original bound; Lemma 4.2, Lemma 5.1, and Corollary 6.3 with their preceding construction and hypotheses for the refinement. https://arxiv.org/html/2610.15483v1.

[2] T. Tao, *An experimental AI-math challenge: using tenth discriminant bounds to explore arguments ruling out Siegel zeroes*, 20 October 2026. Online challenge.